

YouTube Viral Advisor - Demo Script

Demo script - plain text, ~555 words, ~4 minutes at 140 wpm

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Problem Overview

Every YouTuber faces the same question before hitting publish: "Is this video actually going to reach people?"

The problem is that most creators guess - wrong title length, wrong upload time, wrong number of tags. We built YouTube Viral Advisor to turn that guesswork into a data-driven prediction, using real trending data from 11 countries and 178,000 videos collected in 2026.

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Data & Solution

The foundation is a multi-country trending dataset - US, UK, Japan, Brazil, and 7 more. After cleaning, we have 178,267 videos across 15 categories, each enriched with features we engineered ourselves:

Like ratio and comment ratio - proxies for engagement quality.

Title NLP features - character length, word count, ALL-CAPS ratio, presence of numbers, questions, and exclamation marks.

Timing features - publish hour and day of week.

Tag count - how many tags the creator applied.

The Dashboard gives creators an instant read of the landscape. We can see that Pets & Animals dominates with a median of 1.2 million views, that publishing around 13:00 to 21:00 UTC consistently outperforms other windows, and that viral videos cluster in a specific tag count and title length range - which we use in the predictor.

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Viral Advisor Demo

This is the core product. A creator fills in four things they know before uploading: title, category, number of tags, and publish time.

Let's test a real scenario: title is "Top 10 BEST Moments of 2026", category Entertainment, 15 tags, publishing Friday at 15:00 UTC.

The model returns a virality score - a calibrated probability that this video lands in the top 25% of trending videos.

Below the score, we get a Benchmark Comparison table - a side-by-side of the creator's inputs against what viral videos actually look like in the data. Each row has a clear status indicator: green dot for on-target, red for off.

Then Recommendations - specific, data-driven suggestions. Not generic tips, but insights like: "Your title is 34 characters - viral videos use 34 to 66." Or: "Try publishing at 13:00, 15:00, or 21:00 UTC."

The Viral Keywords panel is where NLP comes in. We computed a lift score for every word in the dataset - lift measures how much more likely a word is to appear in viral titles versus average titles. Words already in the title are highlighted. The rest become suggestions.

Finally, a live Views by Publish Hour chart marks the creator's chosen hour in red so they can see exactly where they stand versus the average.

04

Model Insights

The Model Insights page is for full transparency.

The classifier is a Histogram Gradient Boosting model - faster and more accurate than a standard Random Forest on tabular data. It is wrapped in a Calibrated Cross-Validator using isotonic regression, which ensures the percentages we show are statistically meaningful probabilities, not raw scores.

Key results on the held-out test set of 35,000 videos:

ROC AUC: 0.853 - the model correctly ranks a viral video above a non-viral one 85% of the

time.

5-fold CV AUC: 0.797 plus or minus 0.005 - stable across folds, no overfitting.

Brier Score: 0.124 - well-calibrated probabilities.

The permutation feature importance chart shows that category and country are the strongest predictors, followed by tag count, title length, and publish hour - which aligns with intuition and validates the feature engineering.

05

Closing

YouTube Viral Advisor turns a creator's pre-upload decisions into a single, explainable score - backed by 178,000 real trending videos, a calibrated ML model, and NLP keyword analysis.

The goal is not to predict the future perfectly. The goal is to give creators one honest data point before they publish - so the next video is not a guess.